

HR ANALYTICS & EMPLOYEE ATTRITION

ANALYSIS REPORT

1. Project Overview

This project analyses workforce attrition patterns using the IBM HR Analytics Employee Attrition dataset. The primary objective is to identify factors affecting employee resignations and uncover workforce trends related to overtime, salary, age groups, departments, and job satisfaction.

The analysis was performed using the IBM HR Analytics Employee Attrition & Performance Dataset, a widely used HR analytics dataset containing workforce demographics, salary details, job satisfaction metrics, and employee attrition information for 1,470 employees.

2. Dataset Summary

Dataset Information

- Total Records: **1,470 employees**
- Dataset: **IBM HR Analytics Employee Attrition & Performance**
- Tools Used:
 - Excel
 - PostgreSQL
 - Power BI

Important Features

- Employee Demographics:
 - Age
 - Gender
 - Marital Status
 - Education Field
- Work-related Features:
 - Department
 - Job Role
 - Overtime
 - Job Satisfaction

- Monthly Income
- Years at Company
- Attrition Indicators:
 - Attrition Status
 - Salary Slab
 - Work-Life Balance
 - Performance Rating

3. Data Cleaning & Preparation using Excel

The dataset was cleaned and transformed before analysis.

Steps Performed

1. Checked for duplicate employee records.
2. Verified data consistency and formatting.
3. Converted numerical and categorical columns into proper data types.
4. Created:
 - Age Group column
 - Salary Slab column
5. Standardized column names for analysis.
6. Prepared structured data for SQL integration and dashboard development.

4. Data Analysis using PostgreSQL

Structured SQL analysis was performed to answer important HR business questions.

Key Analysis Performed

1. Department-wise Attrition

Research & Development recorded the highest employee attrition, followed by Sales.

	department character varying (50) 	attrition_count bigint 
1	Research & Developm...	133
2	Sales	92
3	Human Resources	12

2. Overtime Impact on Attrition

Employees working overtime showed higher resignation rates compared to employees not working overtime.

	overtime character varying (10) 🔒	employees_left bigint 🔒
1	No	110
2	Yes	127

3. Attrition by Age Group

Employees aged 26–35 contributed the highest employee turnover.

	age_group text 🔒	attrition_count bigint 🔒
1	26-35	116
2	18-25	44
3	36-45	43
4	46+	34

4. Salary Slab Analysis

Employees with lower salary slabs experienced significantly higher attrition.

	salary_slab text 🔒	attrition_count bigint 🔒
1	Medium Sala...	49
2	High Salary	25
3	Low Salary	163

5. Gender-wise Attrition

Male employees contributed a larger percentage of overall employee resignations.

	gender character varying (20) 🔒	total_employees bigint 🔒
1	Female	588
2	Male	882

6. Job Satisfaction Analysis

Lower job satisfaction levels were associated with higher employee attrition.

	jobsatisfaction integer	attrition_count bigint
1	1	66
2	2	46
3	3	73
4	4	52

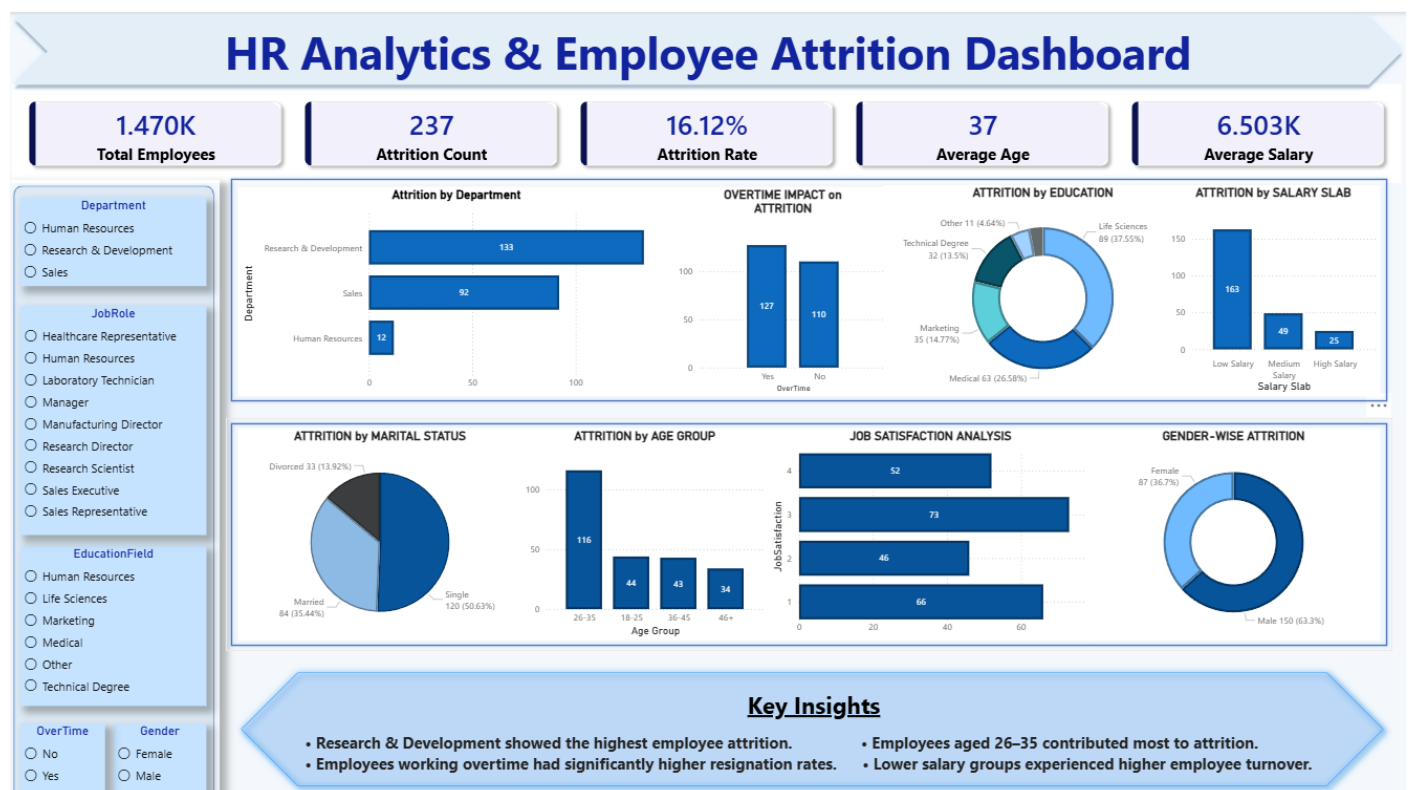
7. Marital Status Analysis

Single employees showed higher attrition compared to married employees.

	maritalstatus character varying (20)	attrition_count bigint
1	Married	84
2	Divorced	33
3	Single	120

5. Dashboard Development in Power BI

An interactive HR analytics dashboard was developed to visualize workforce trends and employee attrition patterns.



Dashboard Features

- KPI Cards:
 - Total Employees
 - Attrition Count
 - Attrition Rate
 - Average Age
 - Average Salary
- Interactive Visuals:
 - Attrition by Department
 - Overtime Impact on Attrition
 - Attrition by Education Field
 - Attrition by Salary Slab
 - Gender-wise Attrition
 - Job Satisfaction Analysis
 - Attrition by Age Group
 - Attrition by Marital Status
- Interactive Slicers:
 - Department
 - Job Role
 - Education Field
 - Overtime
 - Gender

6. Key Insights

- Overall employee attrition rate was **16.12%**.
- Research & Development department recorded the highest employee turnover.
- Employees working overtime showed significantly higher resignation rates.
- Employees aged 26–35 experienced the highest attrition.

- Low salary groups were more likely to leave the organization.
- Male employees contributed around 63% of total attrition.
- Employees with lower job satisfaction levels showed higher turnover trends.

7. Business Recommendations

- Reduce excessive overtime workload to improve employee retention.
- Introduce employee engagement and wellness initiatives.
- Improve compensation strategies for low salary groups.
- Develop retention programs for employees aged 26–35.
- Conduct regular employee satisfaction surveys.
- Strengthen career growth opportunities and internal promotions.